

 YTU – Faculty of Chemical and Metallurgical Engineering. Questions and Answers Sheet	NOTE CHART				
	Q1-Q2	Q3	Q4	Q5	Total
Student Name and Surname					
Students Number					
Course Name	Theory of Linear Programming (1 st midterm)			Exam Date	23/11/2023
Course Instructor Name and Surname	Assoc. Prof. Dr. Hale Gonce Köçken			Signature	
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)					

Q1) (10 p) A carpenter makes tables and chairs. Each table can be sold for a profit of \$30 and each chair for a profit of \$10. The carpenter can afford to spend up to 40 hours per week working and takes six hours to make a table and three hours to make a chair. If necessary, additional workers can be hired at an additional cost of \$5 per hour. Customer demand requires that he makes at least three times as many chairs as tables. Tables take up four times as much storage space as chairs and there is room for at most four tables each week. Formulate this problem as a linear programming problem.

 YTU – Faculty of Chemical and Metallurgical Engineering. Questions and Answers Sheet	NOTE CHART				
	Q1-Q4	Q5	Q6	Q7	Total
Student Name and Surname					
Students Number					
Course Name	Theory of Linear Programming (1 st midterm)			Exam Date	23/11/2023
Course Instructor Name and Surname	Dr. Gökhan GÖKSU			Signature	
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Q1. 5p. In a linear programming problem, when the objective function is parallel to one of the constraints, then:

- a. the solution is not optimal. b. multiple optimal solutions may exist.
c. a single corner point solution exists. d. no feasible solution exists. e. None

$\max Z = x_1 + x_2$ $-2x_1 + x_2 = 3$ $4x_1 + 5x_2 \geq 4$ $3x_1 - 2x_2 \leq 5$ $2x_1 + 7x_2 \leq 7$ $2x_1 - 3x_2 \geq 6$ $x_1, x_2 \geq 0$	Q2. 15p. Consider the problem on the left. In the starting simplex <u>table</u>: The total number of variables <u>is</u>: a. 5 b. 7 c. 8 <u>d. 9</u> e. 10 The total number of slack variables <u>is</u>: a. 0 b. 1 <u>c. 2</u> d. 3 e. 4 The total number of surplus variables <u>is</u>: a. 0 b. 1 <u>c. 2</u> d. 3 e. 4 The total number of artificial variables <u>is</u>: a. 0 b. 1 c. 2 <u>d. 3</u> e. 4 The objective function value at the starting basic feasible solution is _____
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Q3. The optimal simplex table for the given problem is on the right.

$$\max Z = 60x_1 + 35x_2 + 20x_3$$

$$8x_1 + 6x_2 + x_3 \leq 48$$

$$4x_1 + 2x_2 + 1.5x_3 \leq 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 48$$

$$x_2 \leq 5$$

$$x_1, x_2, x_3 \geq 0$$

i. 10p. Which of the following (x_1, x_2, x_3) is NOT an optimal solution to the problem?

a. $(1, 4/5, 48/5)$ b. $(0, 8/5, 56/5)$

c. $(2, 0, 8)$ **d. $(1, 20/3, 1)$**

ii. 5p. Write all the optimal solutions.

Basic	x_1	x_2	x_3	S_1	S_2	S_3	S_4	Sol.
Z	0	0	0	0	10	10	0	280
S_1	0	-2	0	1	2	-8	0	24
x_3	0	-2	1	0	2	-4	0	8
x_1	1	5/4	0	0	-1/2	3/2	0	2
S_4	0	1	0	0	0	0	1	5
Z								

All optimal solution
 $(2\alpha, \frac{8}{5} - \frac{8\alpha}{5}, \frac{56}{5} - \frac{36\alpha}{5})$ for $\alpha \in [0, 1]$

Q4. 5p. A maximization linear programming problem has been solved by the simplex method and the final simplex table is:

Basic	x	y	z	u	v	w	Solution
Objective	0	0	0	2	0	3	7
y	1	1	0	5	0	0	8
x	0	0	0	-3	1	1	3
z	1	0	1	1	0	4	2

Which of the following (x, y, z) is an optimal point indicated by the table?

a. $(8, 3, 2)$ b. $(8, 2, 3)$ **c. $(0, 8, 2)$**

d. $(0, 0, 2)$ e. None

I wish you success.

Q5. 15p. A carpenter makes tables and chairs. Each table can be sold for a profit of \$30 and each chair for a profit of \$10. The carpenter can afford to spend up to 40 hours per week working, and it takes six hours to make a table and three hours to make a chair. If necessary, additional workers can be hired at an additional cost of \$5 per hour. Customer demand requires that he make at least three times as many chairs as tables. For storage, there is space for up to four tables each week. Tables take up four times as much storage space as chairs. Formulate this problem as a linear programming problem.

Q6. 15p. Solve the given problem with tabular form of simplex method.

$$\max Z = 3x_1 + 2x_2 + 3x_3$$

$$2x_1 + x_2 + x_3 \leq 2$$

$$3x_1 + 4x_2 + 2x_3 \geq 8$$

$$x_1, x_2, x_3 \geq 0$$

$$\max Z = 4x_1 + 3x_2$$

$$x_1 + x_2 \leq 12$$

$$2x_1 + x_2 \leq 16$$

$$x_1 \geq 0, x_2 \geq 0$$

Q7. Solve the given linear programming problem with

a. 10p. Graphical method

b. 10p. Algebraic method (not the algebraic form of simplex!)

c. 10p. Simplex method with tabular form.